

## Chapter 4: Score-Based Perspective

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Deep Generative Models Notes

Energy-Based Models

Score-Based Models

Denoising Score Matching



### 1 Energy-Based Models

#### Definition 1: Energy-Based Model (EBM)

Let  $x \in \mathbb{R}^D$  be a data point. An energy-based model defines the density

$$p_\phi(x) = \frac{1}{Z_\phi} \exp(-E_\phi(x)), \quad Z_\phi = \int \exp(-E_\phi(x)) dx.$$

Here  $E_\phi : \mathbb{R}^D \rightarrow \mathbb{R}$  is the energy function parameterized by a neural network.

The lower the energy, the higher the probability. Since all probability mass must sum to one, lowering one region's energy necessarily redistributes mass globally across the whole space.

#### 1.1 Why MLE Is Intractable in EBM

For data distribution  $p_{\text{data}}$ , maximum likelihood learning minimizes

$$\begin{aligned} \mathcal{L}_{\text{MLE}}(\phi) &= - \mathbb{E}_{x \sim p_{\text{data}}} [\log p_\phi(x)] \\ &= \mathbb{E}_{x \sim p_{\text{data}}} [E_\phi(x)] + \log \int \exp(-E_\phi(x)) dx. \end{aligned}$$

The first term decreases energies on real data, while the second term is the normalization penalty. In high dimension,  $\log Z_\phi$  is generally intractable.

## 1.2 Score Function

### Definition 2: Score Function

For a density  $p(x)$ , its score function is

$$s(x) \triangleq \nabla_x \log p(x), \quad s : \mathbb{R}^D \rightarrow \mathbb{R}^D.$$

The vector  $s(x)$  points toward directions of higher probability.

### Remark 1: Why Score Is Useful

Suppose  $p(x) = \tilde{p}(x)/Z$  with constant  $Z = \int \tilde{p}(x) dx$ . Then

$$\nabla_x \log p(x) = \nabla_x \log \tilde{p}(x) - \nabla_x \log Z = \nabla_x \log \tilde{p}(x).$$

So the score does not depend on the normalization constant.

### Proposition 1: Complete Representation via Score

Let  $x_0$  be a reference point. If  $s(x) = \nabla_x \log p(x)$ , then

$$\log p(x) = \log p(x_0) + \int_0^1 s(x_0 + t(x - x_0))^\top (x - x_0) dt.$$

*Proof.* Define the line  $\gamma(t) = x_0 + t(x - x_0)$  for  $t \in [0, 1]$ , so  $\gamma(0) = x_0$ ,  $\gamma(1) = x$ , and  $\gamma'(t) = x - x_0$ . By the chain rule,

$$\frac{d}{dt} \log p(\gamma(t)) = \nabla_x \log p(\gamma(t))^\top \gamma'(t) = s(\gamma(t))^\top (x - x_0).$$

Integrating both sides over  $t \in [0, 1]$  and applying the fundamental theorem of calculus to the left-hand side,

$$\log p(x) - \log p(x_0) = \int_0^1 \frac{d}{dt} \log p(\gamma(t)) dt = \int_0^1 s(x_0 + t(x - x_0))^\top (x - x_0) dt,$$

which is the claimed identity.  $\square$

This identity shows that modeling the score is as expressive as modeling the density itself.

## 1.3 Training via Score Matching

The score matching objective is

$$\mathcal{L}_{\text{SM}}(\phi) \triangleq \frac{1}{2} \mathbb{E}_{x \sim p_{\text{data}}} \|\nabla_x \log p_\phi(x) - \nabla_x \log p_{\text{data}}(x)\|_2^2.$$

Although  $p_{\text{data}}$  is unknown, one can use the tractable Hyvärinen form:

### Proposition 2: Hyvärinen's Identity (Energy-Based Form)

$$\mathcal{L}_{\text{SM}}(\phi) = \mathbb{E}_{x \sim p_{\text{data}}} \left[ -\text{tr}(\nabla_x^2 E_\phi(x)) + \frac{1}{2} \|\nabla_x E_\phi(x)\|_2^2 \right] + C,$$

where  $C = \frac{1}{2} \mathbb{E}_{x \sim p_{\text{data}}} \|\nabla_x \log p_{\text{data}}(x)\|_2^2$  does not depend on  $\phi$ .

*Proof.* Expand the square in the score matching objective:

$$\mathcal{L}_{\text{SM}}(\phi) = \frac{1}{2} \mathbb{E}_{x \sim p_{\text{data}}} \|\nabla_x \log p_\phi(x)\|_2^2 - \mathbb{E}_{x \sim p_{\text{data}}} [\nabla_x \log p_\phi(x)^\top \nabla_x \log p_{\text{data}}(x)] + C,$$

with  $C$  as above. For the cross term, since  $p_{\text{data}}(x) \nabla_x \log p_{\text{data}}(x) = \nabla_x p_{\text{data}}(x)$ ,

$$\mathbb{E}_{x \sim p_{\text{data}}} [\nabla_x \log p_\phi(x)^\top \nabla_x \log p_{\text{data}}(x)] = \int \nabla_x \log p_\phi(x)^\top \nabla_x p_{\text{data}}(x) dx = \sum_{i=1}^D \int \partial_i \log p_\phi(x) \partial_i p_{\text{data}}(x) dx.$$

Assuming  $p_{\text{data}}(x) \rightarrow 0$  as  $\|x\| \rightarrow \infty$  so that boundary terms vanish, integrate each term by parts in  $x_i$ :

$$\int \partial_i \log p_\phi(x) \partial_i p_{\text{data}}(x) dx = - \int p_{\text{data}}(x) \partial_i^2 \log p_\phi(x) dx.$$

Summing over  $i$ ,

$$\mathbb{E}_{x \sim p_{\text{data}}} [\nabla_x \log p_\phi(x)^\top \nabla_x \log p_{\text{data}}(x)] = - \mathbb{E}_{x \sim p_{\text{data}}} [\text{tr}(\nabla_x^2 \log p_\phi(x))].$$

Since  $\log p_\phi(x) = -E_\phi(x) - \log Z_\phi$ , we have  $\nabla_x^2 \log p_\phi(x) = -\nabla_x^2 E_\phi(x)$ , so the cross term equals  $\mathbb{E}_{x \sim p_{\text{data}}} [\text{tr}(\nabla_x^2 E_\phi(x))]$ . Substituting back,

$$\mathcal{L}_{\text{SM}}(\phi) = \frac{1}{2} \mathbb{E}_{x \sim p_{\text{data}}} \|\nabla_x E_\phi(x)\|_2^2 - \mathbb{E}_{x \sim p_{\text{data}}} [\text{tr}(\nabla_x^2 E_\phi(x))] + C,$$

which is the claimed identity.  $\square$

This removes the partition function and avoids model sampling in training, but computing  $\text{tr}(\nabla_x^2 E_\phi(x))$  is still expensive in high dimension.

## 1.4 Langevin Sampling

Given an energy model, discrete-time Langevin dynamics is

$$x_{n+1} = x_n - \eta \nabla_x E_\phi(x_n) + \sqrt{2\eta} \xi_n, \quad \xi_n \sim \mathcal{N}(0, \mathbf{I}).$$

Using  $\nabla_x \log p_\phi(x) = -\nabla_x E_\phi(x)$ , this is equivalently

$$x_{n+1} = x_n + \eta \nabla_x \log p_\phi(x_n) + \sqrt{2\eta} \xi_n.$$

As  $\eta \rightarrow 0$ , it converges to the Langevin SDE

$$dx(t) = \nabla_x \log p_\phi(x(t)) dt + \sqrt{2} dw(t),$$

where  $w(t)$  is standard Brownian motion. The update above is exactly its Euler–Maruyama discretization.

### Remark 2: Challenge

In high-dimensional, multi-modal distributions with many isolated modes, Langevin dynamics can mix very slowly and may spend long time in only a subset of high-probability regions.

## 2 From Energy-Based to Score-Based Models

### Definition 3: Direct Score Modeling

Instead of parameterizing an energy, we directly learn a neural score network

$$s_\phi(x) \approx s(x) \triangleq \nabla_x \log p_{\text{data}}(x), \quad x \in \mathbb{R}^D.$$

The objective is

$$\mathcal{L}_{\text{SM}}(\phi) \triangleq \frac{1}{2} \mathbb{E}_{x \sim p_{\text{data}}} \|s_\phi(x) - s(x)\|_2^2.$$

### 2.1 Tractable Score Matching

A tractable objective with the same minimizer is given by the following generalization of Proposition 2 to an arbitrary vector field  $s_\phi$  (not necessarily a gradient of a scalar potential).

### Proposition 3: Hyvärinen's Identity (General Score Form)

$$\tilde{\mathcal{L}}_{\text{SM}}(\phi) = \mathbb{E}_{x \sim p_{\text{data}}} \left[ \text{tr}(\nabla_x s_\phi(x)) + \frac{1}{2} \|s_\phi(x)\|_2^2 \right]$$

satisfies

$$\mathcal{L}_{\text{SM}}(\phi) = \tilde{\mathcal{L}}_{\text{SM}}(\phi) + C,$$

where  $C = \frac{1}{2} \mathbb{E}_{x \sim p_{\text{data}}} \|s(x)\|_2^2$  is independent of  $\phi$ .

*Proof.* Expand the square in  $\mathcal{L}_{\text{SM}}(\phi) = \frac{1}{2} \mathbb{E}_{x \sim p_{\text{data}}} \|s_\phi(x) - s(x)\|_2^2$ :

$$\mathcal{L}_{\text{SM}}(\phi) = \frac{1}{2} \mathbb{E}_{x \sim p_{\text{data}}} \|s_\phi(x)\|_2^2 - \mathbb{E}_{x \sim p_{\text{data}}} [s_\phi(x)^\top s(x)] + C.$$

Since  $s(x) = \nabla_x \log p_{\text{data}}(x)$  and  $p_{\text{data}}(x) \nabla_x \log p_{\text{data}}(x) = \nabla_x p_{\text{data}}(x)$ ,

$$\mathbb{E}_{x \sim p_{\text{data}}} [s_\phi(x)^\top s(x)] = \int s_\phi(x)^\top \nabla_x p_{\text{data}}(x) \, dx = \sum_{i=1}^D \int s_{\phi,i}(x) \partial_i p_{\text{data}}(x) \, dx,$$

where  $s_{\phi,i}$  is the  $i$ -th component of  $s_\phi$ . Assuming  $p_{\text{data}}(x) \rightarrow 0$  as  $\|x\| \rightarrow \infty$  so boundary terms vanish, integrate each term by parts in  $x_i$ :

$$\int s_{\phi,i}(x) \partial_i p_{\text{data}}(x) \, dx = - \int p_{\text{data}}(x) \partial_i s_{\phi,i}(x) \, dx.$$

Summing over  $i$  gives  $\sum_i \partial_i s_{\phi,i}(x) = \text{tr}(\nabla_x s_\phi(x))$  (the divergence of  $s_\phi$ ), so

$$\mathbb{E}_{x \sim p_{\text{data}}} [s_\phi(x)^\top s(x)] = - \mathbb{E}_{x \sim p_{\text{data}}} [\text{tr}(\nabla_x s_\phi(x))].$$

Substituting back,

$$\mathcal{L}_{\text{SM}}(\phi) = \mathbb{E}_{x \sim p_{\text{data}}} \left[ \text{tr}(\nabla_x s_\phi(x)) + \frac{1}{2} \|s_\phi(x)\|_2^2 \right] + C = \tilde{\mathcal{L}}_{\text{SM}}(\phi) + C.$$

This matches Proposition 2 in the special case  $s_\phi(x) = -\nabla_x E_\phi(x)$ , where  $\text{tr}(\nabla_x s_\phi(x)) = -\text{tr}(\nabla_x^2 E_\phi(x))$ .  $\square$

Therefore the optimizer obeys

$$s^*(x) = \nabla_x \log p(x).$$

So we do not need access to the true score value explicitly.

## 2.2 Sampling with Learned Score

After training, sample with

$$x_{n+1} = x_n + \eta s_{\phi^*}(x_n) + \sqrt{2\eta} \xi_n, \quad \xi_n \sim \mathcal{N}(0, \mathbf{I}),$$

which is the Euler–Maruyama discretization of

$$dx(t) = s_{\phi^*}(x(t)) dt + \sqrt{2} dw(t),$$

with initialization  $x(0)$  from an initial distribution.

## 3 Denoising Score Matching

The tractable score matching loss of Section 2 still requires  $\text{tr}(\nabla_x s_{\phi}(x))$ , which is expensive to compute in high dimension. We first introduce sliced score matching, then a more robust alternative: denoising score matching.

### 3.1 Sliced Score Matching

#### Fact 1: Random Projection of the Trace

Let  $u \in \mathbb{R}^d$  be an isotropic random vector satisfying

$$\mathbb{E}[u] = 0, \quad \mathbb{E}[uu^\top] = \mathbf{I}.$$

Then for any matrix  $A$ ,

$$\text{tr}(A) = \mathbb{E}_u [u^\top A u], \quad \mathbb{E}_u [(u^\top s_{\phi}(x))^2] = \|s_{\phi}(x)\|_2^2.$$

*Proof.* Write  $u^\top A u = \sum_{i,j} u_i A_{ij} u_j$ . Taking expectation and using linearity,

$$\mathbb{E}_u [u^\top A u] = \sum_{i,j} A_{ij} \mathbb{E}[u_i u_j] = \sum_{i,j} A_{ij} \delta_{ij} = \sum_i A_{ii} = \text{tr}(A),$$

where  $\delta_{ij}$  is the Kronecker delta, using  $\mathbb{E}[uu^\top] = \mathbf{I}$ , i.e.  $\mathbb{E}[u_i u_j] = \delta_{ij}$ . For the second identity, apply the first with  $A = s_{\phi}(x) s_{\phi}(x)^\top$ :

$$\mathbb{E}_u [(u^\top s_{\phi}(x))^2] = \mathbb{E}_u [u^\top s_{\phi}(x) s_{\phi}(x)^\top u] = \text{tr}(s_{\phi}(x) s_{\phi}(x)^\top) = s_{\phi}(x)^\top s_{\phi}(x) = \|s_{\phi}(x)\|_2^2.$$

□

Consequently,

$$\mathcal{L}_{\text{SSM}}(\phi) = \mathbb{E}_{x,u} \left[ u^\top (\nabla_x s_{\phi}(x)) u + \frac{1}{2} (u^\top s_{\phi}(x))^2 \right],$$

which can be computed with automatic differentiation.

### Remark 3: Interpretation

This means we only check the model along a random direction: the projected score is judged to align with regions of higher data density, so data points become stationary in expectation.

## 3.2 Denoising Score Matching

Sliced score matching still relies on the raw data distribution, and so we have a more robust alternative: denoising score matching. The most naive objective function is

$$\mathcal{L}_{\text{SM}}(\phi) = \frac{1}{2} \mathbb{E}_{x \sim p_{\text{data}}} \|s_{\phi}(x) - \nabla_x \log p_{\text{data}}(x)\|_2^2.$$

**(1) Vincent's Solution.** Inject noise into  $x \sim p_{\text{data}}$ , using  $p_{\sigma}(\tilde{x} | x)$  with scale  $\sigma$ . The neural network  $s_{\phi}(\tilde{x}; \sigma)$  is trained to approximate the score of the marginal perturbed distribution,

$$p_{\sigma}(\tilde{x}) = \int p_{\sigma}(\tilde{x} | x) p_{\text{data}}(x) dx,$$

by minimizing the loss

$$\mathcal{L}_{\text{SM}}(\phi; \sigma) = \frac{1}{2} \mathbb{E}_{\tilde{x} \sim p_{\sigma}} \|s_{\phi}(\tilde{x}; \sigma) - \nabla_{\tilde{x}} \log p_{\sigma}(\tilde{x})\|_2^2.$$

$\nabla_{\tilde{x}} \log p_{\sigma}(\tilde{x})$  is generally intractable. Conditioning on  $x \sim p_{\text{data}}$  produces an equivalent, tractable objective function:

$$\mathcal{L}_{\text{DSM}}(\phi; \sigma) = \frac{1}{2} \mathbb{E}_{x \sim p_{\text{data}}} \mathbb{E}_{\tilde{x} \sim p_{\sigma}(\cdot | x)} \|s_{\phi}(\tilde{x}; \sigma) - \nabla_{\tilde{x}} \log p_{\sigma}(\tilde{x} | x)\|_2^2.$$

### Theorem 1: Equivalence of $\mathcal{L}_{\text{SM}}$ and $\mathcal{L}_{\text{DSM}}$

The objectives  $\mathcal{L}_{\text{SM}}(\phi; \sigma)$  and  $\mathcal{L}_{\text{DSM}}(\phi; \sigma)$  share the same optimal minimizer:

$$s^*(\tilde{x}; \sigma) = \nabla_{\tilde{x}} \log p_{\sigma}(\tilde{x}).$$

*Proof.* Fix  $\sigma$  and write  $s_{\phi} = s_{\phi}(\cdot; \sigma)$  for brevity. Expanding the squared norms,

$$\mathcal{L}_{\text{SM}}(\phi; \sigma) = \frac{1}{2} \mathbb{E}_{\tilde{x} \sim p_{\sigma}} \|s_{\phi}(\tilde{x})\|_2^2 - \mathbb{E}_{\tilde{x} \sim p_{\sigma}} [s_{\phi}(\tilde{x})^{\top} \nabla_{\tilde{x}} \log p_{\sigma}(\tilde{x})] + C_1,$$

$$\mathcal{L}_{\text{DSM}}(\phi; \sigma) = \frac{1}{2} \mathbb{E}_{x \sim p_{\text{data}}} \mathbb{E}_{\tilde{x} \sim p_{\sigma}(\cdot | x)} \|s_{\phi}(\tilde{x})\|_2^2 - \mathbb{E}_{x \sim p_{\text{data}}} \mathbb{E}_{\tilde{x} \sim p_{\sigma}(\cdot | x)} [s_{\phi}(\tilde{x})^{\top} \nabla_{\tilde{x}} \log p_{\sigma}(\tilde{x} | x)] + C_2,$$

where  $C_1, C_2$  collect the  $\phi$ -independent quadratic terms in  $\nabla_{\tilde{x}} \log p_{\sigma}(\tilde{x})$  and  $\nabla_{\tilde{x}} \log p_{\sigma}(\tilde{x} | x)$  respectively. Since  $p_{\sigma}(\tilde{x}) = \int p_{\sigma}(\tilde{x} | x) p_{\text{data}}(x) dx$  is exactly the marginal of the joint  $p_{\text{data}}(x) p_{\sigma}(\tilde{x} | x)$ , the quadratic terms agree:

$$\mathbb{E}_{x \sim p_{\text{data}}} \mathbb{E}_{\tilde{x} \sim p_{\sigma}(\cdot | x)} \|s_{\phi}(\tilde{x})\|_2^2 = \mathbb{E}_{\tilde{x} \sim p_{\sigma}} \|s_{\phi}(\tilde{x})\|_2^2.$$

It remains to match the cross terms. Differentiating  $p_{\sigma}(\tilde{x}) = \int p_{\sigma}(\tilde{x} | x) p_{\text{data}}(x) dx$  with respect to  $\tilde{x}$ ,

$$\nabla_{\tilde{x}} p_{\sigma}(\tilde{x}) = \int \nabla_{\tilde{x}} p_{\sigma}(\tilde{x} | x) p_{\text{data}}(x) dx = \int p_{\sigma}(\tilde{x} | x) \nabla_{\tilde{x}} \log p_{\sigma}(\tilde{x} | x) p_{\text{data}}(x) dx,$$

using  $\nabla_{\tilde{x}} p_{\sigma}(\tilde{x} | x) = p_{\sigma}(\tilde{x} | x) \nabla_{\tilde{x}} \log p_{\sigma}(\tilde{x} | x)$ . Dividing by  $p_{\sigma}(\tilde{x})$  and using Bayes' rule  $p_{\text{data}}(x | \tilde{x}) = p_{\sigma}(\tilde{x} | x) p_{\text{data}}(x) / p_{\sigma}(\tilde{x})$ ,

$$\nabla_{\tilde{x}} \log p_{\sigma}(\tilde{x}) = \int p_{\text{data}}(x | \tilde{x}) \nabla_{\tilde{x}} \log p_{\sigma}(\tilde{x} | x) dx = \mathbb{E}_{x \sim p_{\text{data}}(\cdot | \tilde{x})} [\nabla_{\tilde{x}} \log p_{\sigma}(\tilde{x} | x)].$$

Hence

$$\begin{aligned} \mathbb{E}_{\tilde{x} \sim p_{\sigma}} [s_{\phi}(\tilde{x})^{\top} \nabla_{\tilde{x}} \log p_{\sigma}(\tilde{x})] &= \mathbb{E}_{\tilde{x} \sim p_{\sigma}} \left[ s_{\phi}(\tilde{x})^{\top} \mathbb{E}_{x \sim p_{\text{data}}(\cdot | \tilde{x})} [\nabla_{\tilde{x}} \log p_{\sigma}(\tilde{x} | x)] \right] \\ &= \mathbb{E}_{\tilde{x} \sim p_{\sigma}} \mathbb{E}_{x \sim p_{\text{data}}(\cdot | \tilde{x})} [s_{\phi}(\tilde{x})^{\top} \nabla_{\tilde{x}} \log p_{\sigma}(\tilde{x} | x)] \\ &= \mathbb{E}_{x \sim p_{\text{data}}} \mathbb{E}_{\tilde{x} \sim p_{\sigma}(\cdot | x)} [s_{\phi}(\tilde{x})^{\top} \nabla_{\tilde{x}} \log p_{\sigma}(\tilde{x} | x)], \end{aligned}$$

where the last line uses that  $p_{\sigma}(\tilde{x}) p_{\text{data}}(x | \tilde{x}) = p_{\text{data}}(x) p_{\sigma}(\tilde{x} | x)$  is the same joint density of  $(x, \tilde{x})$  viewed two ways. This is exactly the cross term of  $\mathcal{L}_{\text{DSM}}(\phi; \sigma)$ . Therefore

$$\mathcal{L}_{\text{SM}}(\phi; \sigma) = \mathcal{L}_{\text{DSM}}(\phi; \sigma) + (C_1 - C_2),$$

with  $C_1 - C_2$  independent of  $\phi$ , so the two objectives share the same minimizer  $s^*(\tilde{x}; \sigma) = \nabla_{\tilde{x}} \log p_{\sigma}(\tilde{x})$ .  $\square$

#### Remark 4: Conditioning Technique

This conditioning technique also appears in DDPM, and in flow-based perspectives (e.g. flow matching).

#### Example 1: Gaussian Noise Injection

Consider injecting Gaussian noise:

$$\tilde{x} = x + \sigma \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \mathbf{I}) \implies p_{\sigma}(\tilde{x} | x) = \mathcal{N}(\tilde{x}; x, \sigma^2 \mathbf{I}).$$

So the conditional score is

$$\nabla_{\tilde{x}} \log p_{\sigma}(\tilde{x} | x) = \frac{x - \tilde{x}}{\sigma^2}.$$

Hence the DSM loss simplifies to

$$\mathcal{L}_{\text{DSM}}(\phi; \sigma) = \frac{1}{2} \mathbb{E}_{x, \tilde{x} | x} \left\| s_{\phi}(\tilde{x}; \sigma) - \frac{x - \tilde{x}}{\sigma^2} \right\|_2^2 = \frac{1}{2} \mathbb{E}_{x, \varepsilon} \left\| s_{\phi}(x + \sigma \varepsilon; \sigma) + \frac{\varepsilon}{\sigma} \right\|_2^2, \quad \varepsilon \sim \mathcal{N}(0, \mathbf{I}).$$

When  $\sigma$  is small, the Gaussian-smoothed marginal  $p_{\sigma} = p_{\text{data}} * \mathcal{N}(0, \sigma^2 \mathbf{I})$  has a score close to that of the clean distribution:  $\nabla_{\tilde{x}} \log p_{\sigma}(\tilde{x}) \approx \nabla_x \log p_{\text{data}}(x)$ . A small step along  $\nabla_{\tilde{x}} \log p_{\sigma}(\tilde{x})$  then moves a noisy sample toward high-probability regions of  $p_{\text{data}}$ . This is the same idea as in score matching. When  $\sigma$  is large, the smoothing over-simplifies the landscape. In practice, DSM uses small noise levels.

### 3.3 Sampling

Let  $s_{\phi^*}(\tilde{x}; \sigma)$  be the trained model. Use Langevin dynamics to replace the true score with the learned model to generate samples. Update rule:

$$\tilde{x}_{n+1} = \tilde{x}_n + \eta s_{\phi^*}(\tilde{x}_n; \sigma) + \sqrt{2\eta} \xi_n, \quad \xi_n \sim \mathcal{N}(0, \mathbf{I}), \quad n = 0, 1, 2, \dots$$

starting from an initial value  $\tilde{x}_0$ . When  $\eta$  is small, after many iterations  $\tilde{x}_n$  approaches samples from  $p_{\text{data}}$ .

**Remark 5: Advantages**

Noise lifts data off a lower-dimensional manifold, giving  $p_\sigma$  full support on  $\mathbb{R}^d$ , so  $\nabla_{\tilde{x}} \log p_\sigma(\tilde{x})$  is well defined everywhere. The noise also smooths out the sparse regions between modes, enhancing training signal quality and facilitating Langevin dynamics to traverse low-density regions more effectively.

### 3.4 Why DSM Is Denoising: Tweedie's Formula

Assume a Gaussian-corrupted observation  $\tilde{x} \sim \mathcal{N}(\cdot; \alpha x, \sigma^2 \mathbf{I})$  from an unknown  $x \sim p_{\text{data}}$ , with  $\alpha \neq 0$ . The noisy marginal is defined as

$$p_\sigma(\tilde{x}) \triangleq \int \mathcal{N}(\tilde{x}; \alpha x, \sigma^2 \mathbf{I}) p_{\text{data}}(x) dx.$$

**Theorem 2: Tweedie's Formula**

Suppose  $x \sim p_{\text{data}}$  and  $\tilde{x} \sim \mathcal{N}(\cdot; \alpha x, \sigma^2 \mathbf{I})$  with  $\alpha \neq 0$ . Then

$$\alpha \mathbb{E}_{x \sim p(x|\tilde{x})} [x | \tilde{x}] = \tilde{x} + \sigma^2 \nabla_{\tilde{x}} \log p_\sigma(\tilde{x}),$$

where the expectation is over the posterior distribution  $p(x | \tilde{x})$  of  $x$  given  $\tilde{x}$ .

*Proof.* Write  $q(\tilde{x} | x) = \mathcal{N}(\tilde{x}; \alpha x, \sigma^2 \mathbf{I})$ , so  $p_\sigma(\tilde{x}) = \int q(\tilde{x} | x) p_{\text{data}}(x) dx$ . Since  $q(\tilde{x} | x) \propto \exp(-\|\tilde{x} - \alpha x\|_2^2 / (2\sigma^2))$ ,

$$\nabla_{\tilde{x}} q(\tilde{x} | x) = -\frac{\tilde{x} - \alpha x}{\sigma^2} q(\tilde{x} | x).$$

Differentiating under the integral sign,

$$\nabla_{\tilde{x}} p_\sigma(\tilde{x}) = \int \nabla_{\tilde{x}} q(\tilde{x} | x) p_{\text{data}}(x) dx = -\frac{1}{\sigma^2} \int (\tilde{x} - \alpha x) q(\tilde{x} | x) p_{\text{data}}(x) dx.$$

Dividing by  $p_\sigma(\tilde{x})$  and using Bayes' rule  $p(x | \tilde{x}) = q(\tilde{x} | x) p_{\text{data}}(x) / p_\sigma(\tilde{x})$ ,

$$\nabla_{\tilde{x}} \log p_\sigma(\tilde{x}) = -\frac{1}{\sigma^2} \int (\tilde{x} - \alpha x) p(x | \tilde{x}) dx = -\frac{1}{\sigma^2} \left( \tilde{x} - \alpha \mathbb{E}_{x \sim p(\cdot|\tilde{x})} [x | \tilde{x}] \right).$$

Rearranging,

$$\alpha \mathbb{E}_{x \sim p(\cdot|\tilde{x})} [x | \tilde{x}] = \tilde{x} + \sigma^2 \nabla_{\tilde{x}} \log p_\sigma(\tilde{x}),$$

which is the claimed formula.  $\square$

A denoised estimate is obtained by nudging  $\tilde{x}$  a step of size  $\sigma^2$  in the direction of  $\nabla_{\tilde{x}} \log p_\sigma(\tilde{x})$ . In other words, Tweedie's formula lets us obtain clean samples/denoised estimates from noisy observations.